**THE PRESIDENT'S OFFICE**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**
**NAMANGA SECONDARY SCHOOL**

**FORM FOUR PRE-MOCK EXAMINATION**
**031 PHYSICS**

**Time: 3 Hours** **2025**

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**Instructions**

1. This paper consists of sections **A, B** and **C** with total of eleven (11) questions.
2. Answer all questions from section **A** and **B** and any two (2) questions from section **C.**
3. Where necessary, the following constants may be used
   - ❖ Acceleration due to gravity = $10ms^{-2}$.
   - ❖ Density of water = $1.0gcm^{-3}$ or $1000kgm^{-3}$

**SECTION A (16 MARKS)**
**Answer all questions**

1. For each of the items (i) – (x), choose the correct answer from the given alternatives and write its letter beside the item number in the booklet provided.

(i) Which of the following best describes the primary goal of physics as a science?
   A. To create complex mathematical equations
   B. To describe and understand natural phenomena
   C. To conduct chemical experiments
   D. To manufacture technological devices
   E. To deal with how fundamental laws work

(ii) Which of the following factors does NOT affect the pressure exerted by a gas in a closed container?
   A. Temperature of the gas
   B. Volume of the container
   C. Number of gas molecules
   D. Color of the gas
   E. None of the above

(iii) According to Archimedes' principle, what determines the buoyant force acting on a submerged object in water?
   A. The weight of the object
   B. The volume of water displaced by the object
   C. The size of the object
   D. The density of the object
   E. Density of fluid

(iv) Colors are produced when light passes through a glass prism because:
   A. Light waves interfere
   B. Glass prism colours the light
   C. In glass, different colurs travel at different speed
   D. Different colurs are filtered
   E. Diffraction of light occurs

(v) The focal point of a concave mirror is
   A. A point where parallel rays of light converge
   B. A point where parallel rays of light diverge
   C. The center of the mirror
   D. Always located behind the mirror
   E. The point in the concave mirror which is very difficult to determine

(vi) What does relative humidity measure?
   A. The amount of moisture in the air compared to the maximum amount it can hold at a certain temperature
   B. The total amount of water vapor present in the atmosphere
   C. The temperature at which air becomes saturated with moisture
   D. The density of water vapor in the air
   E. The temperature at which the air becomes saturated with water vapor

(vii) Which layer of the earth is primarily composed of molten rock?
   A. Crust
   B. Mantle
   C. Inner core
   D. Outer core
   E. No correct answer

(viii) What is the main reason for the changing phases of the Moon?
   A. Earth's rotation
   B. The distance between Earth and the Moon
   C. The reflection of sunlight on the Moon
   D. The changing distance of the Moon from the Earth
   E. Force gravity between the moon and the earth

(ix) What phenomenon causes a prism to separate white light into its component colors?
   A. Reflection
   B. Refraction
   C. Diffraction
   D. Dispersion
   E. Relegation

(x) In a series circuit, if one component fails, what happens to the current flow?
   A. It continues to flow uninterrupted
   B. The current flow increases
   C. The current flow decreases
   D. The current flow stops
   E. The current flows for sometimes the stops

2. Match the items in list A with responses in list B by writing the letter of the correct response beside.

| LIST A | LIST B |
|---|---|
| i. Specific heat capacity<br>ii. Heat capacity<br>iii. Latent heat of fusion<br>iv. Latent heat<br>v. Regelation<br>vi. Evaporation | A. The energy possessed by object due to its temperature<br>B. The phenomena of melting and re-freezing under the influence of pressure<br>C. The process through which the liquid changes to vapor (gas) at a temperature even below its boiling point.<br>D. It is the vaporization process where liquid changes into gas system when heated<br>E. Refers to the energy absorbed or released by a substance during a phase change, such as melting, freezing, boiling, or condensing, without any change in temperature<br>F. The amount of heat energy required to raise the temperature of one-unit mass of a substance by one degree Celsius (or one Kelvin).<br>G. The amount of energy required to change a substance from solid to liquid (or vice versa) at a constant temperature.<br>H. The total amount of heat energy required to change the temperature of an entire object or system by one degree Celsius (or Kelvin). |

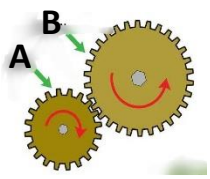
## SECTION B (54 MARKS)

**Answer all questions in this section**

3. (a) The dispersed beam of light (seven colours) can recombine to form single white light. Using two glass prisms, sketch the diagram to show how this is formed.
   (b) A rod of length 10 cm long lies along the principal axis of a concave mirror of focal length 10 cm in such a way that its end closer to the pole is 20 cm away from the mirror. What is the length of the image?

4. (a)(i) A hole at the bottom of a ship is more dangerous than one near the surface. Explain why?
   (ii) Briefly explain, why heavy trucks are made up with many tyres?
   (b) A uniform rod with a mass of 120g and length 130 cm is suspended by a wire from a point 80 cm from the rod's left end.
   (i) What mass must be hung from the right end of the rod for it to be in equilibrium?
   (ii) What will be the tension in the wire?

5. (a) (i) Apply Newton's first law of motion to explain why an object pulled along the ground with a constant velocity has zero net force but force on it is not zero.
   (b) A gear wheel A has 20 teeth. It is used to drive a gear wheel be B with 80 teeth.

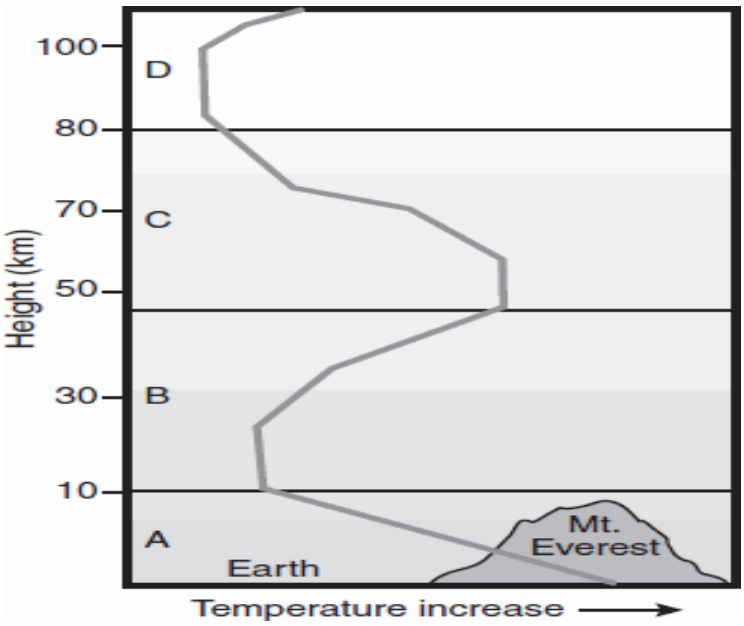



   (i) Calculate the velocity ratio
   (ii) If wheel A rotates three times every second, how many times does wheel B rotates in a second.

6. (a) A beaker containing water ice is heated from -5$^0$C to 0$^0$C and then from 0$^0$C to 15$^0$C. With the aid of a diagram, explain the variation of density with temperature.
   (b) A heater of 500w is used to raise the temperature of 50 kg of material of specific heat capacity of 960 J/kg K, from 18$^0$C to 38$^0$C. Assume that all the heat from the heater is given to the material, calculate:
   (i) Heat capacity of the material
   (ii) The time taken in seconds

7. (a) (i) The half-life of a certain radioactive substance is 64 days. Explain the meaning of this statement.
   (ii) The half-life of $60_{C_o}$ is 5 years. How much time does it take for $(\frac{3}{4})^{th}$ of its initial mass to disintegrate?
   (b) Thorium disintegrates in the following manners
   $90^{Th^{232}} .\rightarrow 88^{Ra^{228}} \rightarrow 89^{Ac^{228}} .\rightarrow 90^{Th^{228}} \rightarrow 88^{Ra^{224}}$. State the particle emitted at each part of the disintegration.

8. (a) State heliocentric theory and outline three shortfalls of the theory.
   (b) Study the vertical structure of the atmosphere in the figure below and then answer the asked questions.




   (i) What is the name of the layer A? The temperature of this layer decreases with an altitude at average rate of 6.5$^0$C/km. What is the reason for this decrease.
   (ii) Why there is an increase in temperature at layer B. What is the name of this layer?
   (iii) Why layer D is called communication layer?

**SECTION C (30 MARKS)**

**Answer only two (2) questions from this section**

9. (a) (i) With support of diagram, what will be the effect on electron as it passes between two plates A and B, one of which carrying a + ve charge and the other carrying a –ve charge? Remember electron carries negative charge.
   (ii) Explain why it is more dangerous to walk barefooted during lightning storms?
   (b) If the resistance is doubled and the potential difference across it is reduced to one third, what is the

ratio of old current to new current?

(c)Each of two new dry cells has an e.m.f of 1.5volts and internal resistance of 1.0 ohm. The two cells are connected in parallel and the combination connected to a 10 ohm resistor. Find the current and heat developed per second on the 10 ohm resistor**.**

10. (a) Explain why is a semiconductor damaged by a strong current.

(b) The collector current $I_C$ in a common emitter amplifier is directly proportional to the base current $I_B$.

(i) What does this statement mean?

(ii)Write down the mathematical expression for the statement.

(iii) If current gain $\beta$ of this amplifier is 200, calculate the collector current given that the base current $I_B$ is $2\mu$A.

(b) Explain why ultraviolet light causes a negatively-charged electroscope to discharge, but not a positively-charged one**.**

11. (a)A helicopter is hovering at an altitude of 200m above the surface of a lake. A speaker on the helicopter is sending out sound waves, which are reflected from both the surface of water and bottom of the lake. If the difference in arrival times of the two echoes is measured to be 0.24 seconds, what is the depth of lake? (The atmospheric temperature is $20^0$C).

(b) (i) Draw a labeled diagram of a simple d.c. motor

(ii) Explain why the coil will rotate when the current flows in it and why it will turn in a specific direction**.**

(iii) Why does such motor occasionally not start to rotate when the current is switched on but will give continuous rotation after a slight push?